ANDROID PROGRAMMING PRACTICAL 05

Module 1 · Fragment-based UI navigation between multiple screens

Aim

Implement Fragment-based UI navigation between multiple screens.

Project name: FragmentNavigationDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create MainActivity and two fragments, HomeFragment and DetailsFragment, with separate XML layouts.

2. Place a FragmentContainerView in activity_main.xml with a stable ID.

3. Add a button in HomeFragment to open DetailsFragment.

4. Use parentFragmentManager.beginTransaction().replace(containerId, new DetailsFragment()).addToBackStack(null).commit() for navigation.

5. Add a back/navigation control or use the device Back action to return to HomeFragment.

6. Run the app and verify the fragment content changes without launching a new Activity.

7. Test Back navigation and check that the previous fragment is restored.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

Users can navigate between at least two fragment screens and return using Back.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.